

A Novel Herbal Formula Induces Cell Cycle Arrest and Apoptosis in Association With Suppressing the PI3K/AKT Pathway in Human Lung Cancer A549 Cells

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Abstract

Aim of the study. In recent years, the incidence of lung cancer, as well as the mortality rate from this disease, has increased. Moreover, because of acquired drug resistance and adverse side effects, the effectiveness of current therapeutics used for the treatment of lung cancer has decreased significantly. Chinese medicine has been shown to have significant antitumor effects and is increasingly being used for the treatment of cancer. However, as the mechanisms of action for many Chinese medicines are undefined, the application of Chinese medicine for the treatment of cancer is limited. The formula tested has been used clinically by the China National Traditional Chinese Medicine Master, Professor Zhonging Zhou for treatment of cancer. In this article, we examine the efficacy of Ke formula in the treatment of non-small cell lung cancer and elucidate its mechanism of action. **Methods.** A Balb/c nude mouse xenograft model using A549 cells was previously established. The mice were randomly divided into normal, mock, Ke, cisplatin (DDP), and co-formulated (Ke + DDP) groups. After 15 days of drug administration, the animals were sacrificed, body weight and tumor volume were recorded, and the tumor-inhibiting rate was calculated. A cancer pathway finder polymerase chain reaction array was used to monitor the expression of 88 genes in tumor tissue samples. The potential antiproliferation mechanism was also investigated by Western blot analysis. **Results.** Ke formula minimized chemotherapy-related weight loss in tumor-bearing mice without exhibiting distinct toxicity. Ke formula also inhibited tumor growth, which was associated with the downregulation of genes in the PI3K/AKT, MAPK, and WNT/ β -catenin pathways. The results from Western blot analyses further indicated that Ke blocked the cell cycle progression at the G1/S phase and induced apoptosis mainly via the PI3K/AKT pathway. **Conclusion.** Ke formula inhibits tumor growth in an A549 xenograft mouse model with no obvious side effects. Moreover, Ke exhibits synergistic antitumor effects when combined with DDP. The mechanism of action of Ke is to induce cell cycle arrest and apoptosis by suppressing the PI3K/AKT pathway. Further research will be required to determine the mechanism of action behind the synergistic effect of Ke and DDP.

Keywords

Zhou's formula of Ke Jin Yan, cisplatin, non-small cell lung cancer, cell cycle, apoptosis, PI3K/AKT pathway

Introduction

Non-small cell lung cancer (NSCLC) accounts for approximately 85% of lung cancer cases.^{1,2} Although lung cancer can be treated using various therapies, including chemotherapy and radiotherapy, these treatments often have adverse side effects.^{3–5} As such, new drugs with fewer negative side effects are needed. A potential strategy for the treatment of NSCLC is to use medicinal plants that are effective against lung cancer but have fewer side effects.

For many years, traditional Chinese medicine (TCM) has been a prime source of anticancer treatments. However, a single herb is seldom used alone. Herbal formulas use

synergy and interactions between a myriad phytochemicals present in different herbs to achieve therapeutic efficacy by targeting multiple biological and pathological processes

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while minimizing adverse side effects.⁶⁻⁹ One such formula, Ke formula, has been used to treat cancer for many years. Indeed, clinical observations from studies performed using Ke formula suggest that this formula has some efficacy as a cancer treatment.¹⁰⁻¹² Therefore this formula is of sufficient interest to merit laboratory testing to further investigate its antitumor properties and elucidate its mechanism of action.

In this study, we used polymerase chain reaction (PCR) microarrays to investigate the activity of Ke formula in murine xenograft models to examine gene expression. Based on the microarray data, we performed an in-depth study of potential Ke formula-mediated signaling pathways involved in the regulation of cell growth and apoptosis.

Materials and Methods

Cell Line and Culture Conditions

Human lung carcinoma (NCI-A549) cells obtained from the Cell Line Bank (Shanghai, China) were used for the experiments. The cell line was cultured in RPMI-1640 (Gibco, Carlsbad, CA) with 10% fetal bovine serum (Gibco, Carlsbad, CA), 100 U/mL penicillin, and 100 mg/mL streptomycin (Amresco) in a 5% CO₂ cell culture incubator (Forma, Thermo Scientific, Waltham, MA) at 37°C.

Drugs

Ke formula is made up of 13 herbs, *Semen Benincasae* (15 g), *Arisaema Cum Bile* (12 g), *Houttuynia Cordata* (15 g), *Rhizoma Fagopyri Dibotryis* (15 g), *Cremastra Appendiculata* (10 g), *Herba et Gemma Agrimoniae* (15 g), *Carthami Flos* (10 g), *Lignum Sappan* (10 g), *Rhizoma Pinelliae* (10 g), *Radix Ranunculi Ternati* (15 g), *Radix Adenophorae* (12 g), *Radix Ophiopogonis* (12 g), and *Rhizoma Polygonati Odorati* (12 g), according to the conventional method. The table in the appendix lists the names of these herbs in Chinese, English, and Latin.

The criteria for the quality of the herbs used were in accordance with the 2005 edition of Chinese pharmacopoeia (Chinese Pharmacopoeia Commission, Pharmacopoeia of People's Republic of China, Beijing: People's Medical Publishing House; 2005). Before use in experiments, the herbs were tested for heavy metals, microbial contamination, and residual pesticides; all results met the safety standards in China. Laboratory personnel were blinded as to the identity of the herbs. A trained technician prepared the decoction according to a standardized procedure. Each unit of formula yielded 1200 mL of decoction. Cisplatin (*cis*-diamminedichloroplatinum [DDP]) was purchased from Nanjing Pharmaceutical Factory, Ltd.

Experimental Animals

Balb/c athymic (nude) mice (female, age 6-8 weeks, n = 40) weighing 20 ± 2 g were purchased from the Animal

Center in the Academy of Military Medical Sciences (Beijing, China). The animal room was controlled for temperature (22°C ± 2°C), light (12-hour light/dark cycle), and humidity (50% ± 10%). All laboratory feed pellets and bedding were autoclaved.

The study was approved by the Jiangsu Animal Care and Use Committee and followed national and institutional rules regarding animal experiments.

Xenograft Experiments

A nude mouse model of tumor regression was used to evaluate suppression of solid tumor growth in mice treated with different drugs. A total of 1 × 10⁷ A549 cells in 0.2 mL culture medium were subcutaneously injected into the axilla of each mouse. After 7 days of observation, solid tumor masses were excised from the NCI-A549-inoculated mice. When the tumor volume reached approximately 50 mm³, the mice were randomly distributed into 4 groups of 8: mock (saline alone), Ke (1.8 g/kg/d Ke formula), DDP (5 mg/kg/d DDP), and co-formulated (Ke formula + DDP, same dosage as for individual treatment). Ke formula was orally administered. DDP was peritoneally injected daily for 15 days. Mice with no tumors served as the control group and were orally administered saline. The dosage of Ke formula was converted from the human clinical dosage. The effects of different concentrations of Ke formula on Lewis lung carcinoma cell transplanted tumors and survival conditions of C57/BL6 mice were examined.

Antitumor Activity In Vivo

All animals were monitored for activity, physical condition, bodyweight, and tumor growth. The body weight of each animal was measured once every 2 days. The longest (*a*) and the shortest (*b*) tumor diameters (mm) were obtained, and the formula for an ellipsoid sphere ($0.52 \times a \times b^2$) was used to calculate the tumor volume every 2 days. The tumor weights were also measured on the final day of the experiment after the animals were sacrificed. The antitumor activities of the different treatments were expressed as inhibition rate:

$$\text{Inhibition rate (\%)} = (W_{\text{mock}} - W_{\text{treatment}}) / W_{\text{model}} \times 100\%$$

where W_{mock} and $W_{\text{treatment}}$ are the tumor weights of the mock and treatment groups, respectively. After the weight was measured, the xenograft tumors were frozen in liquid nitrogen for further experiments.

Polymerase Chain Reaction Microarray Analysis

Total RNA was extracted from experimental tumor samples using Trizol reagent (Invitrogen, Carlsbad, CA). RNA was purified using an RNeasy MinElute Cleanup Kit (Qiagen, Valencia, CA) according to the manufacturer's protocol. RNA concentration was quantified by UV absorption using

a NanoDrop spectrophotometer (Bio Lab, Ltd, Jerusalem, Israel). cDNA was synthesized from total RNA with Super Script.III Reverse Transcriptase (Invitrogen, Carlsbad, CA), and used for real-time PCR with the lung cancer pathway finder PCR array (Cat. No. PA-112, Super Array, Bioscience Corporation, Foster City, CA). The $\Delta\Delta\text{Ct}$ method was used to analyze the PCR data.

- (a) ΔCt was calculated for each pathway-focused gene in each treatment group.
 $\Delta\text{Ct} (\text{group 1}) = \text{average Ct} - \text{average of HK genes' Ct for group 1 array}$
 $\Delta\text{Ct} (\text{group 2}) = \text{average Ct} - \text{average of HK genes' Ct for group 2 array}$
- (b) $\Delta\Delta\text{Ct}$ was calculated for each gene across 2 PCR arrays (or groups).
 $\Delta\Delta\text{Ct} = \Delta\text{Ct} (\text{group 2}) - \Delta\text{Ct} (\text{group 1})$
- (c) Fold-change was calculated for each gene from groups 1 to 2 as $2^{-\Delta\Delta\text{Ct}}$.

Western Blot Analysis

Cancer tissues were ground in a mortar with liquid nitrogen, lysed in tissue lysis buffer (50 mM Tris (pH 7.4), 150 mM NaCl, 1% Triton X-100, 1% sodium deoxycholate, 0.1% sodium dodecyl sulfate (SDS), and $1 \times$ cocktail containing 0.5 mM Na_3VO_4 , 50 mM NaF, and 5 mM phenylmethyldulfonyl fluoride), then *sonicated* on ice with an ultrasonic disrupter for 30 minutes. The resulting homogenate was centrifuged at 12000 rpm/min for 30 min. After the supernatant was harvested, equal amounts of proteins were separated by 10% SDS–polyacrylamide gel electrophoresis and transferred to a polyvinylidene difluoride membrane (Millipore, Billerica, MA). The membrane was probed with antibodies directed against p-AKT^(ser473), p-AKT^(thr308), AKT, Bax, Bcl-2, CDK4, CyclinD3, E2F1, P21, P27, p-PTEN^(ser380), p-GSK3 β ^(ser9) (all from Cell Signaling Technology, Danvers, MA), and β -actin (Santa Cruz Biotechnology, Santa Cruz, CA). The antibodies were diluted in phosphate-buffered saline (pH 7.4) containing 0.05% (v/v) Tween-20 and 5% (w/v) dried milk. The blots were incubated with the appropriate horseradish peroxidase–conjugated secondary antibodies (Cell Signaling Technology, Danvers, MA). Bound antibodies were visualized by Super Signal West Dura Extended Duration Substrate (Thermo Scientific, Waltham, MA) using a Gel Imaging System (ChemDoc XRS System170-8070, Bio-Rad, Hercules, CA).

Identification of Pathways Using the Kyoto Encyclopedia of Genes and Genomes (KEGG) and BioCarta Databases

Gene identification was performed to determine which biochemical pathways were altered by the different treatments. After identifying genes from the PCR microarray data, we

investigated whether any of these genes were part of the same pathway. To this end, the KEGG (<http://www.genome.ad.jp/kegg/pathway.html>) and BioCarta (<http://www.biocarta.com/genes/allpathways.asp>) biochemical pathway databases were searched using the genes selected from the PCR microarray and Western blot analyses.

Statistical Analysis

The data were analyzed using SPSS 11.5 software and presented as mean $\pm$ standard deviation ($\bar{x} \pm \text{SD}$). Significant difference between the means of the mock and treatment groups was analyzed using one-way analysis of variance followed by Dunnett's *t*-test correction and paired *t*-test. Statistical significance was determined at $P < .05$ and $P < .01$.

Results

Ke Suppresses the Growth of NCI-A549 Cells in Mice

Over the course of treatment, mice in the DDP group exhibited significant weight loss and slowed activity (Figure 1A). In contrast, those in the Ke group maintained stable weight and normal activity, and those in the Ke + DDP group exhibited marked improvement (Figure 1A). While all treatments inhibited the growth of NCI-A549-transplanted solid tumors as compared with the mock group (Figure 1B), Ke significantly inhibited tumor growth as compared with the mock group (significant differences in final tumor weight [$P < .05$] and volume [$P < .01$] were observed). After the animals were sacrificed (Figure 1C), the tumor-inhibition rate was 62.8%. After DDP combination, the tumor-inhibition rate was 82.5% (Table 1). These data indicate that Ke formula resulted in minimal chemotherapy-related body weight loss in mice bearing tumors and did not adversely affect activity.

Gene Expression Profiles Vary Among the Different Treatment Groups

The lung cancer pathway finder PCR array was used to analyze changes in gene expression after the treatment and to elucidate the molecular mechanisms by which Ke inhibits tumor growth. A total of 8, 3, and 12 genes exhibited a 2-fold difference in expression after treatment with the different drugs as compared with the mock group. Of these genes, 1, 0, and 2 were upregulated and 7, 3, and 10 were downregulated in the Ke, DDP, and co-formulated groups, respectively (Figure 2). The genes that were differentially expressed are listed and categorized by their putative functions in Tables 2 to 4. In general, the upregulated genes were involved in the TGF β signal pathway, while the genes that were downregulated were involved in cell cycle, apoptosis, proliferation, and angiogenesis belong to the PI3K/AKT, MAPK, and WNT/ β -catenin pathways.

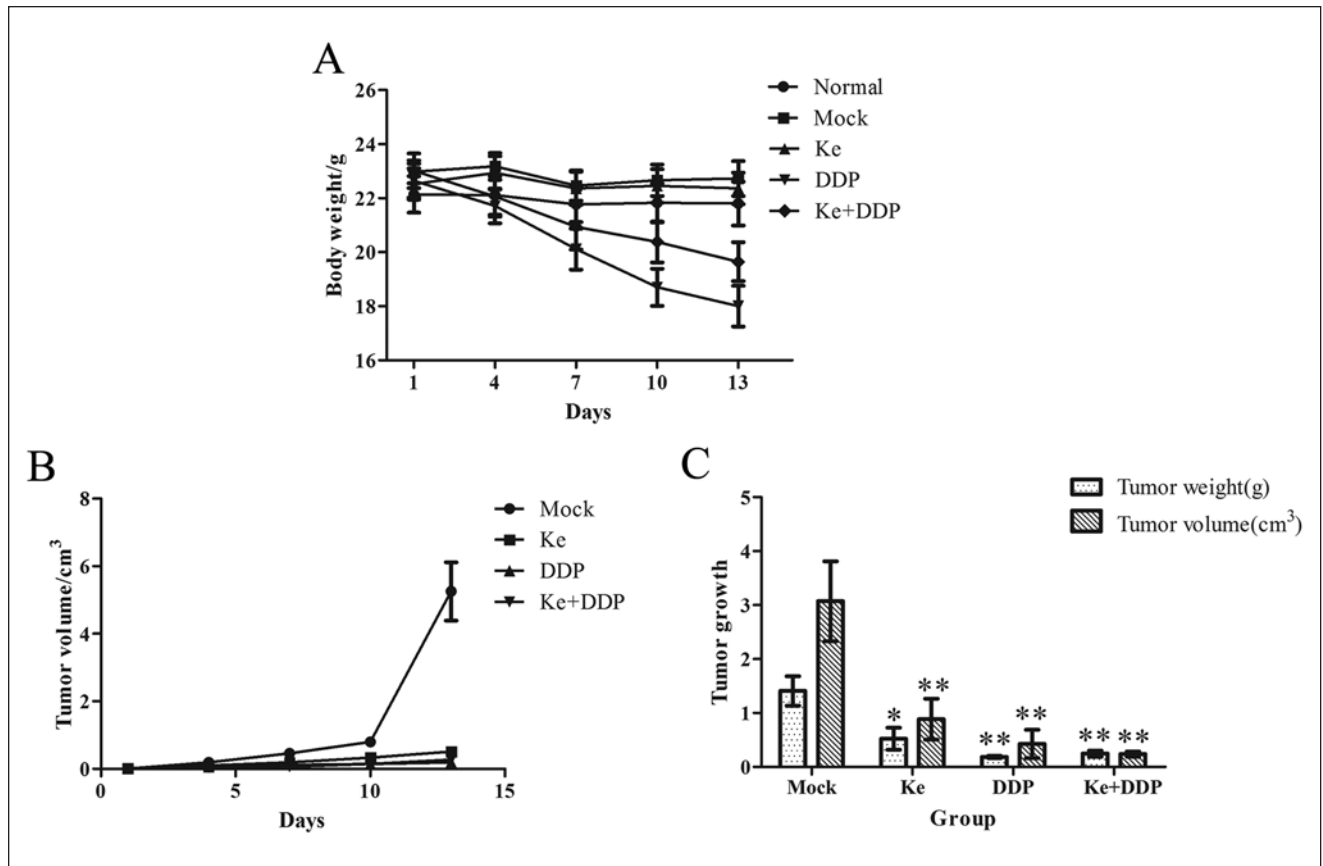


Figure 1. Effects of different drugs on the suppression of NCI-A549 transplanted tumor and survival condition of nude mice. (A) Mice of DDP (Cisplatin) group lost weight significantly, and mice of Ke + DDP group have markedly improved, and all the mice of Ke group keep stable weight with normal activity. (B) All drugs significantly suppress the solid tumor growth compared with the mock group. (C) Ke and DDP significantly inhibited both the volume and weight of tumors, and their combination displays a certain synergy. Values are means of 3 separate experiments $\pm$ standard deviation. * $P < .05$ and ** $P < .01$ by Student's paired *t*-test compared with the mock group.

Table 1. Effects of the Ke on the Tumor Volume and Weight of NCI-A549 Tumor-Bearing Mice.

Group	n	Volume in cm ³ (Mean $\pm$ SD)	Weight in Grams (Mean $\pm$ SD)	Inhibitory Rate (%)
Mock	8	3.07 $\pm$ 1.81	1.41 $\pm$ 0.67	—
Ke	8	1.08 $\pm$ 0.85**	1.01 $\pm$ 0.73*	28.5
DDP	8	0.38 $\pm$ 0.60**	0.18 $\pm$ 0.05**	86.9
Ke + DDP	8	0.37 $\pm$ 0.28**	0.26 $\pm$ 0.20**	81.2

* $P < .05$. ** $P < .01$ versus phosphate-buffered saline control.

Ke Induces Cell Cycle Arrest in the G1/S Phase and Activates Apoptotic Proteins

To predict the putative signaling pathway(s) affected by treatment with Ke, we used the KEGG and BioCarta pathway databases to analyze the differentially expressed profiles of drug-induced genes to fit the transduction-signaling map. Two major pathways were downregulated by Ke: the cell cycle regulatory pathway and apoptotic pathway. Our results

indicate that Ke treatment decreased the expression levels of E2F1 (2.08-fold) and CDK4 (1.09-fold), as well as Bcl-2 and AKT (2.08- and 1.18-fold, respectively). Combination of Ke with DDP resulted in further downregulation of these genes; E2F1 (2.4-fold), CDK4 (1.25-fold), Bcl-2 (2.72-fold), and AKT (1.60-fold). These data suggest that Ke inhibits tumor growth by suppressing PI3K/AKT activation and that the combination of Ke and DDP has a synergistic additive effect.

Ke Suppresses PI3K/AKT Pathway Activation

To determine whether Ke can suppress PI3K/AKT pathway activation, Western blot analysis was used to examine changes in PI3K/AKT pathway activity after herbal treatment. Western blot analysis confirmed that Ke increased the ratio of Bax/Bcl-2, induced apoptosis (Figure 3A), and decreased the levels of CDK4, CyclinD3, and E2F1. However, in terms of apoptosis and cell cycle regulation, the expression levels of P21 and P27 did not change (Figure 3B). In addition, Ke also downregulated the levels of p-AKT (ser473), p-AKT (thr308), p-GSK3 β , and p-PTEN (Figure 3C). In terms of

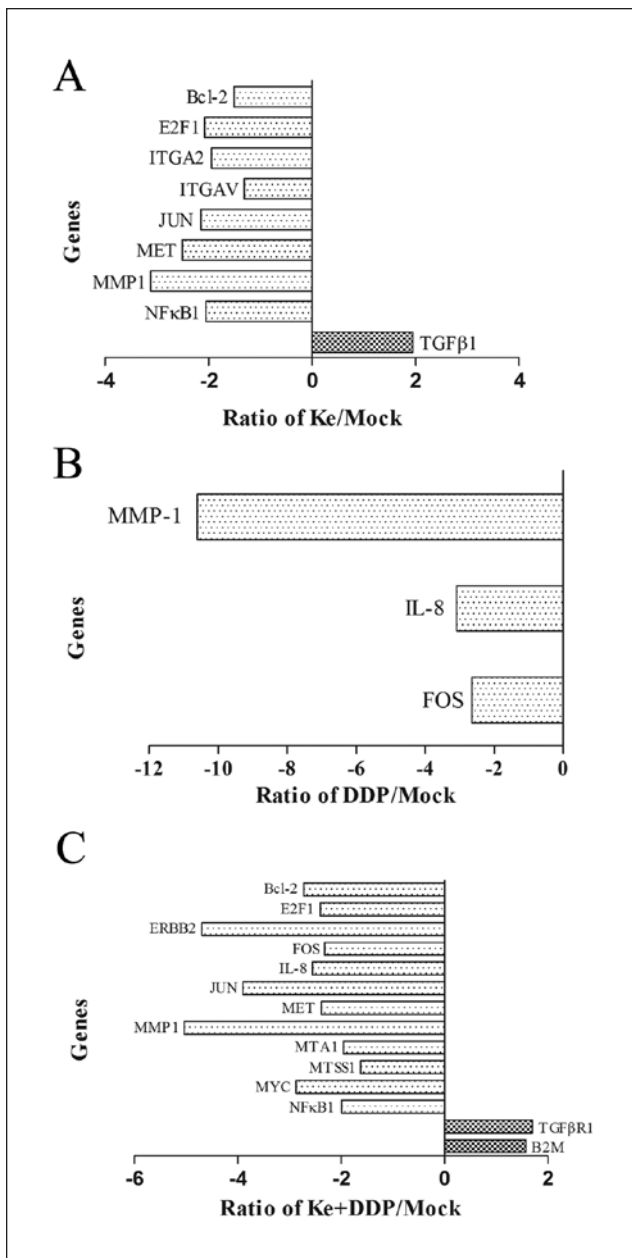


Figure 2. Putative genes expression in differences compared with mock group after drugs treatment. (A) Differential gene expression of Ke. (B) Differential gene expression of DDP (Cisplatin). (C) Differential gene expression of Ke + DDP.

apoptosis and cell cycle, Ke combined with DDP increased the ratio of Bax/Bcl-2, decreased the levels of CDK4 and P27, increased the level of E2F1, and did not affect the levels of CyclinD3 and P21 as compared with the mock group. The levels of p-AKT (ser473), p-AKT (thr308), and p-GSK3β were upregulated, whereas the level of PTEN was not changed in the Ke+DDP group as compared with the mock group. These results indicate that Ke could suppress PI3K/AKT pathway activation, induce apoptosis, arrest the

Table 2. Differential Gene Expression of Ke Compared With Mock.

Symbol	Description	P	Up- or Downregulation
TGFβ1	Transforming growth factor, beta receptor 1	.023768	1.95
NFκB1	Nuclear factor of kappa light polypeptide gene enhancer in B-cells 1	.042290	-2.05
MMP1	Matrix metalloproteinase 1	.010463	-3.12
MET	Met proto-oncogene	.008206	-1.97
JUN	Jun proto-oncogene	.021470	-1.52
ITGAV	Integrin, alpha V	.021102	-2.07
ITGA2	Integrin, alpha 2	.004371	-1.94
E2F1	E2F transcription factor 1	.017151	-2.08
BCL2	B-cell CLL/lymphoma 2	.039776	-1.51

Table 3. Differential Gene Expression of DDP (Cisplatin) Compared With Mock.

Symbol	Description	P	Up- or Downregulation
MMP1	Matrix metalloproteinase 1	.030504	-10.61
IL8	Interleukin 8	.001860	-3.09
FOS	FBJ osteosarcoma oncogene	.029671	-2.65

Table 4. Differential Gene Expression of Ke + DDP Compared With Mock.

Symbol	Description	P	Up- or Downregulation
B2M	Beta-2-microglobulin	.000118	1.58
TGFR1	Transforming growth factor, beta receptor 1	.004611	1.70
NFκB1	Nuclear factor of kappa light polypeptide gene enhancer in B-cells 1	.045609	-1.99
MYC	V-myc myelocytomatosis viral oncogene	.015108	-2.87
MTSS1	Metastasis suppressor 1	.012011	-1.62
MTA1	Metastasis associated 1	.032471	-1.95
MMP1	Matrix metalloproteinase 1	.010746	-5.04
MET	Met proto-oncogene	.001235	-2.38
JUN	Jun proto-oncogene	.004544	-3.90
IL8	Interleukin 8	.006315	-2.55
FOS	FBJ osteosarcoma oncogene	.032380	-2.32
ERBB2	v-erb-b2 erythroblastic leukemia viral oncogene homolog 2	.022632	-4.69
E2F1	E2F transcription factor 1	.000525	-2.40
BCL2	B-cell CLL/lymphoma 2	.002502	-2.72

cell cycle in the G1/S phase, and inhibit the growth of NCI-A549 cell-transplanted solid tumors. Further studies will be required to more precisely define the mechanisms behind the synergistic effect of Ke and DDP.

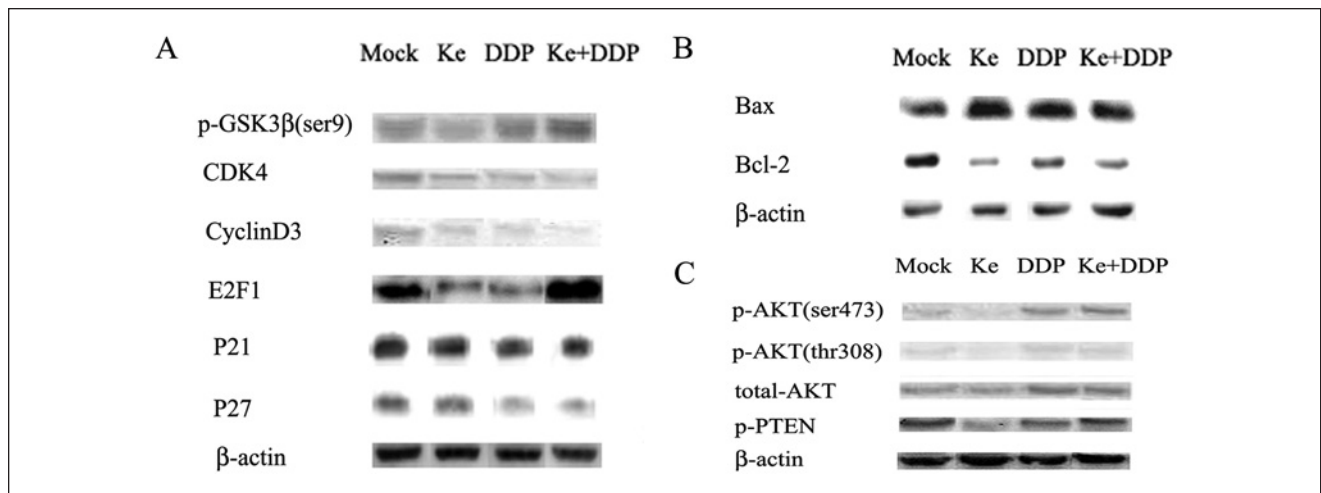


Figure 3. Ke formula induces cell cycle arrest in G1/S phase and activates apoptotic proteins via suppress PI3K/AKT/GSK3 β pathway. (A) The expression levels of cell cycle regulatory proteins were analyzed by Western blotting. (B) The expression levels of apoptosis-related proteins were analyzed by Western blotting. (C) The expression levels of phosphorylation of AKT and PTEN were analyzed by Western blotting.

Discussion

Traditional Chinese medicine provides a rich source of potential chemopreventive and therapeutic agents for the treatment of cancer. Rigorous and systematic evaluations are, however, necessary to establish the efficacy of herbal formulas and transform traditional herbal practices into science-based medicines.¹³⁻¹⁷ In the present article, we evaluated the anticancer activity of Ke formula, which has been shown to be effective in the clinic. The crude formula used in this study followed TCM treatment principles based on syndrome differentiation. TCM theory proposes that lung cancer is mainly related to deficiency of both Qi and Yin, as well as pathological changes of Qi stagnation, blood stasis, and accumulation of phlegm. Therefore, the treatment principle is to resolve phlegm, activate blood and Qi circulation, remove stasis, and nourish Yin. The herbs in Ke formula include herbs of drying dampness and resolving phlegm (*Semen Benincasae*, *Rhizoma Pinelliae*, and *Arisaema Cum Bile*), herbs of relieving masses (*Houttuynia Cordata*, *Rhizoma Fagopyri Dibotryis*, *Cremastra Appendiculata*, and *Radix Ranunculi Ternati*), herbs of activating blood circulation and removing stasis (*Herba et Gemma Agrimoniae*, *Carthami Flos*, and *Lignum Sappan*); and herbs of invigorating Qi and nourishing Yin (*Radix Adenophorae*, *Rhizoma Polygonati Odorati*, and *Radix Ophiopogonis*). In this study, we demonstrated that Ke can inhibit NCI-A549 lung cancer cell growth in vivo and induce cell cycle G1/S arrest and apoptosis. Moreover, the animal study revealed that Ke significantly inhibited the growth of subcutaneous human NCI-A549 xenograft tumors with no obvious side effects. Data from the microarray experiments demonstrated that Ke can upregulate the expression of TGF β , which has a dual function in tumors and could negatively regulate the expression of CDK.¹⁸ In general,

Ke suppressed the expression of genes belonging to certain pathways, including those involved in MAPK signaling, cell cycle progression, and apoptosis. Western blot analysis demonstrated that the key pathway affected was that regulated by activation of AKT. Indeed, our data suggest that Ke affects cell growth mainly by inhibiting the PI3K/AKT pathway.

The PI3K/AKT pathway plays an important role in survival when cancer cells are exposed to different kinds of apoptotic stimuli, such as oxidative and osmotic stresses, irradiation, matrix adhesion, ischemic shock, and chemotherapeutic drugs.^{19,20} Once activated, AKT controls cellular functions, such as apoptosis, cell cycle regulation, gene transcription, and protein synthesis, through the phosphorylation of downstream substrates, such as GSK3 β and Bad. AKT is overexpressed and constitutively active in many human cancers, including NSCLC.²¹⁻²³ The activation of AKT can be blocked by PTEN through inhibition of phosphorylation.^{24,25} In the present study, we evaluated the effect of Ke on the PI3K/AKT pathway. Interestingly, the levels of p-PTEN, p-AKT (ser473), and p-AKT (thr308) decreased after Ke treatment, indicating that Ke may inhibit AKT activation by suppressing PTEN phosphorylation.

Cell cycle regulation is important in growth control, and dysregulation of the cell cycle machinery has been implicated in tumorigenesis.^{26,27} The CDK4/cyclin D complex promotes the progression through the G1 phase into the S phase by phosphorylating retinoblastoma (Rb) protein and releasing E2F1.^{28,29} Overexpression of CDK4 has been described in many tumors, including NSCLC. The activity of CDK/cyclin complexes is negatively regulated by binding to cyclin-dependent kinase inhibitors belonging to the INK4 family (P16, P15, P18, and P19) and the Cip/Kip family (P21, P27, and P57).^{30,31} Among these, P21 induces cell

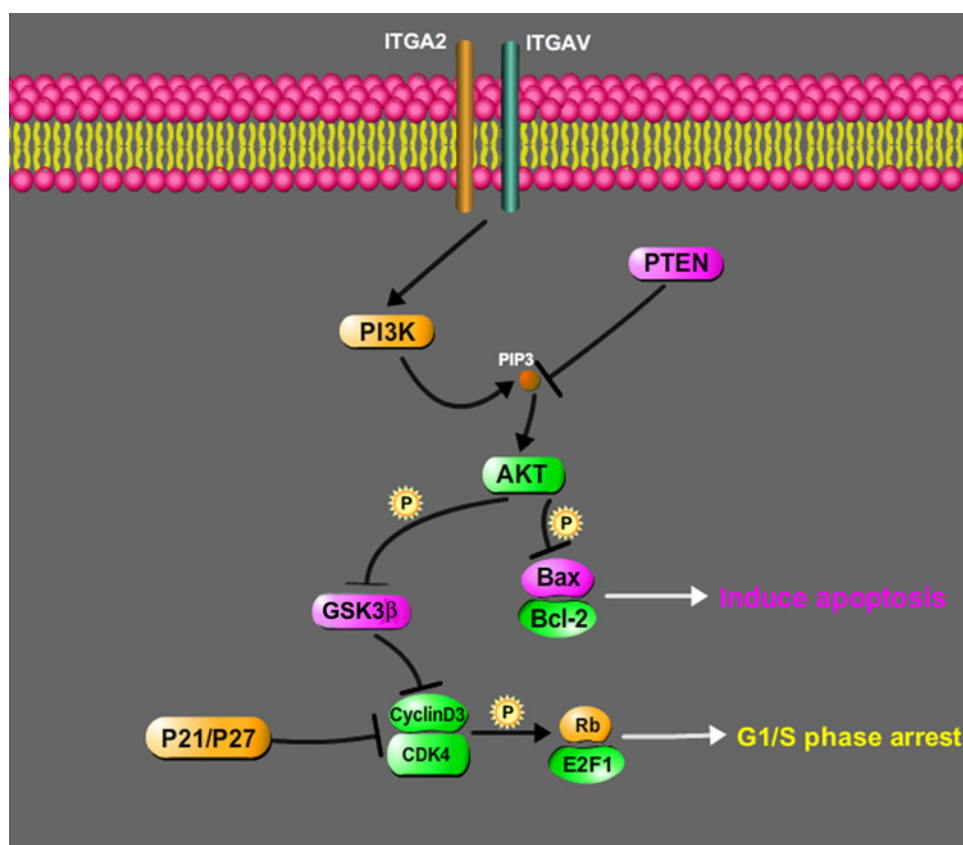


Figure 4. Schematic diagram of Ke-regulated genes in different pathways analyzed by Western blotting and KEGG and BioCarta pathway databases. On regulation of the cell cycle, Ke leads to inactivation of AKT, and hence GSK3 β inhibits the CDK4/CyclinD3 complex, reduces the p-Rb and E2F1's expression, and eventually induces G1/S phase arrest, meanwhile, Ke also increase the expression of Bax and decrease the expression of Bcl-2 to promote apoptosis. Ke-induced genes are colored red, and Ke-suppressed genes are colored green, and those colored yellow mean Ke-unregulated genes.

cycle arrest in the G0/G1 phase by inhibiting the activity of the CDK4/cyclin D complex, and seems to be a fundamental partner of P53.³²⁻³⁴ Our results suggest that the cell cycle could be blocked as the levels of CDK4/CyclinD3 and E2F1 were decreased and those of P21 and P27 did not change after Ke treatment. Therefore, Ke could possibly induce G1/S arrest via a P53-independent mechanism.

Through investigation of signaling pathways, we found another gene-regulating cell cycle process to be important in response to Ke treatment. GSK3 β , as a downstream target of the PI3K/AKT pathway, regulates cell proliferation and differentiation via phosphorylation of a serine residue⁹ by AKT.^{35,36} GSK3 β is a key regulatory enzyme in the WNT/ β -catenin transduction pathway. In addition, it could regulate transcription factors such as JUN and MYC as well as phosphorylate the CDK4/cyclin D complex, leading to rapid ubiquitination,³⁷⁻⁴⁰ degradation of Rb and E2F1, and induction of G1/S arrest.

Apoptosis, an important regulator in developmental processes, maintenance of homeostasis, and elimination of

damaged cells, is the outcome of a complex interaction between pro-apoptotic and anti-apoptotic molecules. Proteins of the Bcl-2 family are key regulators of the apoptotic pathway. The Bcl-2 family can be divided into 2 sub-families: anti-apoptotic proteins such as Bcl-2 and pro-apoptotic proteins such as Bax. Accumulating evidence indicates that many anticancer agents induce apoptosis by targeting the proteins of the Bcl-2 family. The ratio of Bax/Bcl-2 is a critical determinant of whether cells will undergo apoptosis.⁴¹⁻⁴³ In the present study, Ke increased pro-apoptotic Bax expression and decreased anti-apoptotic Bcl-2 expression, leading to an increased Bax/Bcl-2 ratio, which may be one of the molecular mechanisms by which Ke induces apoptosis.

In conclusion, Ke can induce cell cycle arrest at G1/S phase by disrupting regulatory proteins involved in cell cycle progression and initiate apoptosis and by suppressing the expression of genes of focal adhesion and proliferation through PI3K/AKT pathway inactivation (Figure 4). Conversely, the mechanism behind the synergistic effect of

Ke and DDP does not rely on the PI3K/AKT pathway. DDP is a nonspecific cell cycle drug that mainly interferes with DNA replication. Analysis of the microarray data indicates that a close relationship exists between DDP and the MAPK

pathway. Taken together, our results indicate that treatment with Ke alone, or in combination with DDP, would provide therapeutic efficacy against NSCLC, without adverse side effects.

Appendix

Names of Herbs

Chinese	Pinyin	English	Latin
冬瓜子	Dongguazi	Benincasae Semen	Semen Benincasae
法半夏	Fabanzia	Pinellia Tuber	Rhizoma Pinelliae
胆南星	Dannanxing	Bile Arisaema	Arisaema Cun Bile
山慈菇	Shancigu	Pseudobulb of Appendiculate Cremastra	Cremastra Appendiculata
猫爪草	Maozhaocao	Catclaw Buttercup Root	Radix Ranunculi Ternati
金荞麦	Jinqiaomai	Wild Buckwheat Rhizome	Rhizoma Fagopyri Dibotryis
鱼腥草	Yuxingcao	Heartleaf Houttuymia Herb	Houttuymia Cordata
仙鹤草	Xianhecao	Hairyvein Agrimonia Herb and Bud	Herba et Gemma Agrimoniae
苏木	Sumu	Sappan Wood	Lignum Sappan
红花	Honghua	Safflower	Carthami Flos
沙参	Shashen	Fourleaf ladybell Root	Radix Adenophorae
玉竹	Yuzhu	Fragrant solomonseal Rhizome	Rhizoma Polygonati Odorati
麦冬	Maidong	Dwarf Lilyturf Tuber	Radix Ophiopogonis

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Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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References

1. Tamiya M, Kobayashi M, Morimura O, et al. Clinical significance of the serum crosslinked N-telopeptide of type I collagen as a prognostic marker for non-small-cell lung cancer. *Clin Lung Cancer*. 2013;14:50-54.
2. Jemal A, Thomas A, Murray T, Thun M. Cancer statistics, 2002. *CA Cancer J Clin*. 2002;52:23-47.
3. Xu T, Wei Q, Guerra JL, et al. HSPB1 gene polymorphisms predict risk of mortality for US patients after radio(chemo)therapy for non-small cell lung cancer. *Int J Radiat Oncol Biol Phys*. 2012;84:e229-e235.
4. De Ruyscher D, Van Meerbeeck J, Vandecasteele K, et al. Radiation-induced oesophagitis in lung cancer patients: is susceptibility for neutropenia a risk factor? *Strahlenther Onkol*. 2012;188:564-567.
5. Castel M, Pathak A, Despas F, Mazières J. Adverse effects of new biological therapies for non-small-cell bronchial cancer. *Presse Med*. 2011;40(4 pt 1):415-419.
6. Ji XM, Ouyang B, Liu H, et al. In vitro and in vivo inhibitory effect of the combination of Wenxia Changfu formula with cisplatin in non-small cell lung cancer. *Chin J Integr Med*. 2011;17:908-916.
7. Liu J, Hu XJ, Jin B, Qu XJ, Hou KZ, Liu YP. β -Elemene induces apoptosis as well as protective autophagy in human non-small-cell lung cancer A549 cells. *J Pharm Pharmacol*. 2011;64:146-153.
8. Yin H, Guo R, Xu Y, et al. Synergistic antitumor efficiency of docetaxel and curcumin against lung cancer. *Acta Biochim Biophys Sin (Shanghai)*. 2012;44:147-153.
9. Zhang L, Ruan J, Yan L, et al. Xanthatin induces cell cycle arrest at G2/M checkpoint and apoptosis via disrupting NF- κ B pathway in A549 non-small-cell lung cancer cells. *Molecules*. 2012;17:3736-3750.
10. Ji H. *The clinical observation of treatment of later Period non-small cell lung cancer by treated with modified Zhoushi Kejinyan combined with TP cheotherapy* [master's thesis]. Nanjing, China: Nanjing University of Traditional Chinese Medicine; 2011.

11. Lan C, Li L, Zhou Z. Medical master Professor Zhou Zhongying's four experiences in treating cancer. *J Tradit Chin Med*. 2011;5:11-13.
12. Lin S, Wang Z, Zhou Z. Professor Zhou Zhongying's experience on "complex act of generosity" in treating 1 case of lung cancer. *J Henan Tradit Chin Med*. 2010;7:717-718.
13. Wang P, She G, Yang Y, et al. Synthesis and biological evaluation of new ligustrazine derivatives as anti-tumor agents. *Molecules*. 2012;17:4972-4985.
14. Zhang P, Wang X, Xiong S, et al. Genome wide expression analysis of the effect of the Chinese patent medicine Zilongjin tablet on four human lung carcinoma cell lines. *Phytother Res*. 2011;25:1472-1479.
15. Polier G, Ding J, Konkimalla BV, et al. Wogonin and related natural flavones are inhibitors of CDK9 that induce apoptosis in cancer cells by transcriptional suppression of Mcl-1. *Cell Death Dis*. 2011;2:e182.
16. Cheng HM, Li CC, Chen CY, et al. Application of bioactivity database of Chinese herbal medicine on the therapeutic prediction, drug development, and safety evaluation. *J Ethnopharmacol*. 2010;132:429-437.
17. Sigstedt SC, Hooten CJ, Callewaert MC, et al. Evaluation of aqueous extracts of *Taraxacum officinale* on growth and invasion of breast and prostate cancer cells. *Int J Oncol*. 2008;32:1085-1090.
18. Warner BJ, Blain SW, Seoane J, Massague J. Myc downregulation by transforming growth factor beta required for activation of the p15(Ink4b) G(1) arrest pathway. *Mol Cell Biol*. 1999;19:5913-5922.
19. Scrima M, De Marco C, Fabiani F, et al. Signaling networks associated with AKT activation in non-small cell lung cancer (NSCLC): new insights on the role of phosphatidylinositol-3 kinase. *PLoS One*. 2012;7:e30427.
20. Zito CR, Jilaveanu LB, Anagnostou V, et al. Multi-level targeting of the phosphatidylinositol-3-kinase pathway in non-small cell lung cancer cells. *PLoS One*. 2012;7:e31331.
21. Chou PY, Huang GJ, Pan CH, et al. Trilinolein inhibits proliferation of human non-small cell lung carcinoma A549 through the modulation of PI3K/Akt pathway. *Am J Chin Med*. 2011;39:803-815.
22. Acharya BR, Bhattacharyya S, Choudhury D, Chakrabarti G. The microtubule depolymerizing agent naphthazarin induces both apoptosis and autophagy in A549 lung cancer cells. *Apoptosis*. 2011;16:924-939.
23. Espinosa I, Catusus L, Canet B, D'Angelo E, Munoz J, Prat J. Gene expression analysis identifies two groups of ovarian high-grade serous carcinomas with different prognosis. *Mod Pathol*. 2011;24:846-854.
24. Tang JM, He QY, Guo RX, Chang XJ. Phosphorylated Akt overexpression and loss of PTEN expression in non-small cell lung cancer confers poor prognosis. *Lung Cancer*. 2006;51:181-191.
25. Akca H, Demiray A, Aslan M, Acikbas I, Tokgun O. Tumour suppressor PTEN enhanced enzyme activity of GPx, SOD and catalase by suppression of PI3K/AKT pathway in non-small cell lung cancer cell lines. *J Enzyme Inhib Med Chem*. 2013;28:539-544.
26. Lee JW, Min HY, Han AR, et al. Growth inhibition and induction of G1 phase cell cycle arrest in human lung cancer cells by a phenylbutenoid dimer isolated from *Zingiber cassumunar*. *Biol Pharm Bull*. 2007;30:1561-1564.
27. Arora S, Bhardwaj A, Srivastava SK, et al. Honokiol arrests cell cycle, induces apoptosis, and potentiates the cytotoxic effect of gemcitabine in human pancreatic cancer cells. *PLoS One*. 2011;6:e21573.
28. Fedorov SN, Shubina LK, Bode AM, Stonik VA, Dong Z. Dactylone inhibits epidermal growth factor-induced transformation and phenotype expression of human cancer cells and induces G1-S arrest and apoptosis. *Cancer Res*. 2007;67:5914-5920.
29. Mateen S, Tyagi A, Agarwal C, Singh RP, Agarwal R. Silibinin inhibits human nonsmall cell lung cancer cell growth through cell-cycle arrest by modulating expression and function of key cell-cycle regulators. *Mol Carcinog*. 2010;49:247-258. Erratum in *Mol Carcinog*. 2010;49:849.
30. Baldi A, De Luca A, Esposito V, Campioni M, Spugnini EP, Citro G. Tumor suppressors and cell-cycle proteins in lung cancer. *Pathol Res Int*. 2011;2011:605042.
31. Vincenzi B, Schiavon G, Silletta M, et al. Cell cycle alterations and lung cancer. *Histol Histopathol*. 2006;21:423-435.
32. Vonlanthen S, Heighway J, Kappeler A, Altermatt HJ, Borner MM, Betticher DC. p21 is associated with cyclin D1, p16INK4a and pRb expression in resectable non-small cell lung cancer. *Int J Oncol*. 2000;16:951-957.
33. Zhao YF, Wang CR, Wu YM, Ma SL, Ji Y, Lu YJ. P21 (waf1/cip1) is required for non-small cell lung cancer sensitive to Gefitinib treatment. *Biomed Pharmacother*. 2011;65:151-156.
34. Zhao Y, Wu S, Wu J, et al. Introduction of hypoxia-targeting p53 fusion protein for the selective therapy of non-small cell lung cancer. *Cancer Biol Ther*. 2011;11:95-107.
35. Lu Y, Muller M, Smith D, et al. Kinome siRNA-phosphoproteomic screen identifies networks regulating AKT signaling. *Oncogene*. 2011;30:4567-4577.
36. Sabile AA, Arlt MJ, Muff R, et al. Cyr61 expression in Osteosarcoma indicates poor prognosis and promotes intratibial growth and lung metastasis in mice. *J Bone Miner Res*. 2012;27:58-67.
37. Plyte SE, Hughes K, Nikolakaki E, Pulverer BJ, Woodgett JR. Glycogen synthase kinase-3: functions in oncogenesis and development. *Biochim Biophys Acta*. 1992;1114:147-162.
38. Ong CS, Zhou J, Ong CN, Shen HM. Luteolin induces G1 arrest in human nasopharyngeal carcinoma cells via the Akt-GSK-3 β -cyclin D1 pathway. *Cancer Lett*. 2010;298:167-175.
39. Chen B, Pan H, Zhu L, Deng Y, Pollard JW. Progesterone inhibits the estrogen-induced phosphoinositide 3-kinase $\rightarrow$ AKT $\rightarrow$ GSK-3 β $\rightarrow$ cyclin D1 $\rightarrow$ pRB pathway to block uterine epithelial cell proliferation. *Mol Endocrinol*. 2005;19:1978-1990.
40. Hashimoto T, He Z, Ma WY, et al. Caffeine inhibits cell proliferation by G0/G1 phase arrest in JB6 cells. *Cancer Res*. 2004;64:3344-3349.
41. Wu CS, Chen YJ, Chen JJ, et al. Terpinen-4-ol induces apoptosis in human nonsmall cell lung cancer in vitro and in vivo. *Evid Based Complement Alternat Med*. 2012;2012:818261.
42. Wong BS, Hsiao YC, Lin TW, et al. The in vitro and in vivo apoptotic effects of *Mahonia oiwakensis* on human lung cancer cells. *Chem Biol Interact*. 2009;180:165-174.
43. Jackson T, Chougule MB, Ichite N, Patlolla RR, Singh M. Antitumor activity of noscapine in human non-small cell lung cancer xenograft model. *Cancer Chemother Pharmacol*. 2008;63:117-126.